

CSCE 753 COMPUTER VISION · MODULE 05 · LECTURE NOTES

Structure from Motion

Many views, one global solution.

MODULE OUTCOMES

- Explain the structure-from-motion pipeline end to end.
- Explain bundle adjustment and why it is sparse.
- Explain drift and loop closure.
- Distinguish incremental from global approaches.
- Diagnose a reconstruction that has failed.

1 The pipeline

1. FEATURES in every image (Module 03)
2. MATCH candidate image pairs
all pairs is $O(n^2)$ and prohibitive past a few hundred images -- use a vocabulary tree or a learned global descriptor to shortlist candidates
3. GEOMETRIC VERIFICATION per pair (Module 04)
RANSAC an essential matrix; a pair is retained only if it has enough geometrically consistent inliers
4. TRACKS: chain pairwise matches into multi-view observations of a single world point
5. SEED: choose the best-conditioned pair -- wide baseline, many inliers, NOT near-planar, homography test passed -- and reconstruct it
6. LOOP, until no image can be added:
register a new image by PnP against known points
triangulate newly visible points
BUNDLE ADJUST (section 2)
remove points with large residuals
7. FINAL global bundle adjustment

THE SEED CHOICE MATTERS MORE THAN ANYTHING ELSE. A near-degenerate seed pair (Module 04 section 4) poisons every pose built on it, and the symptom shows up fifty images later as a reconstruction that will not converge.

Step 3 is what makes the whole thing robust. A pair whose matches cannot be explained by a single rigid geometry is discarded before it can contaminate anything — **and it is the reason structure from motion works at all on internet photographs**, where a large fraction of candidate pairs are simply unrelated images that happened to share visual words.

PNP: POSE FROM KNOWN POINTS

Once some world points are known, a new image's pose is determined by its 2D observations of them — the perspective-n-point problem, which is the inverse of Module 02's projection with the points known and the camera unknown.

Three points give up to four solutions and a fourth resolves the ambiguity. So P3P inside RANSAC is the standard approach, and the sample size of 3 pays off exponentially in iteration count (Module 03 §3) — another instance of why minimal solvers are worth deriving.

This is how each new image joins the reconstruction, and it is why the incremental pipeline is incremental: each image needs existing structure to register against, so the reconstruction must grow rather than appear at once.

And PnP is the core of camera tracking. A running AR or SLAM system is performing PnP against a map, every frame, at 60 Hz — which means the whole of this module's machinery is what sits behind a phone placing a virtual object on a table, and is the direct bridge from this course to CSCE 650's virtual reality.

2 Bundle adjustment

$$\text{minimise} \quad \sum_{ij} \rho(\| \pi(\mathbf{c}_i, \mathbf{x}_j) - \mathbf{x}_{ij} \|^2)$$

One objective over every camera parameter and every world point simultaneously, summed over all observations, with ρ a robust loss. Thousands of cameras and millions of points means millions of unknowns — **and it is solved routinely, in seconds to minutes**, which is initially surprising.

It is tractable because the Jacobian is extremely sparse. Camera i 's residuals do not involve point j unless camera i actually observed point j , and in any real dataset each camera sees a tiny fraction of the points. **The standard exploitation is the Schur complement:** the points block is block-diagonal (each point is independent of every other, given the cameras), so it can be eliminated analytically, leaving a much smaller dense system in the camera parameters alone. **That reduction is the entire engineering content of a bundle adjustment library**, and it is why Ceres and g2o exist as specialised tools rather than everyone calling a generic optimiser.

WHAT BUNDLE ADJUSTMENT NEEDS TO WORK

A good initialisation. It is a non-convex least-squares problem (CSCE 669 Module 01 §2) and converges to a local minimum — **which is the entire reason the incremental pipeline of §1 exists**. Bundle adjustment is a refinement step, not a solver; something else must get close first.

A robust loss. Plain squared error lets a single mismatched observation dominate the objective, because the residual is quadratic and a gross outlier's residual is enormous. **Huber or Cauchy bounds each observation's influence, and this is not optional** — a pipeline without it fails on real data regardless of how good the matching was.

Gauge fixing. The solution is free up to a global similarity transform — Module 01's scale ambiguity plus an arbitrary global rotation and translation — so **the normal equations are singular with a 7-dimensional null space unless you fix one camera's pose and one distance**. Omitting this produces a solver that wanders or fails to converge for reasons that look numerical and are structural.

And Levenberg–Marquardt rather than plain Gauss–Newton. The damping term is what keeps the step stable when the Jacobian is near-singular, which happens routinely at near-degenerate configurations — it interpolates between Gauss–Newton when the model is trustworthy and gradient descent when it is not, which is CSCE 669 Module 07's trust-region idea in its most-used form.

3 Drift and loop closure

DRIFT IS ACCUMULATED ERROR WITH NO EXTERNAL REFERENCE

Each frame's pose is estimated relative to what came before, so errors compound **multiplicatively rather than averaging out**. A per-frame rotation error of 0.1 degree becomes 36 degrees over 360 frames, and nothing in a purely sequential estimator corrects it.

The result is a reconstruction that is locally excellent and globally bent. A straight corridor comes out curved; a trajectory that returned to its starting point does not close. **Every local measurement is consistent and the global structure is wrong**, which is why residual statistics do not reveal drift.

Loop closure fixes it: recognise that the current frame observes a place already in the map, add the corresponding constraint, and **redistribute the accumulated error over the whole trajectory** by re-optimising with that constraint in place (pose-graph optimisation, which is bundle adjustment over poses alone).

So place recognition is essential infrastructure rather than a feature. And it is an image-retrieval problem — find the database image most similar to this query — which is CSCE 670's subject, and is why bag-of-visual-words indexes and learned global descriptors appear inside SLAM systems that otherwise contain no learning at all.

	Incremental	Global
Method	Add one image at a time, bundle adjusting repeatedly.	Solve for all rotations at once, then all translations, then refine everything.
Robustness	High — a bad image fails to register and is simply rejected, with no effect on what exists.	Lower — one bad pairwise constraint influences the whole rotation-averaging solution.
Cost	Up to $O(n^4)$ in the worst case, because bundle adjustment is repeated as the reconstruction grows.	Substantially cheaper on large image sets, which is what motivated it.
Drift	Accumulates; requires explicit loop closure.	Less drift by construction, since all constraints enter simultaneously.
Used by	COLMAP, and most SLAM front ends.	Internet-scale reconstruction, where n is in the hundreds of thousands.

Incremental wins on robustness and loses on scale, which is why production systems are hybrid: solve global rotations first (which is well-conditioned and cheap), then refine incrementally with the robustness that provides. **The field did not settle on either pure approach**, and that is worth knowing before implementing one.

4 Diagnosing a failed reconstruction

- **The reconstruction splits into disconnected components.** Not enough geometrically verified pairs bridge them. **The fix is capture, not code** — photograph the transition between the two regions.

- **Everything collapses onto a plane or toward a point.** A degenerate seed pair, or a genuinely planar scene (Module 04 §4). **Check the seed pair's homography test first.**
- **A straight corridor comes out curved.** Drift (§3). Needs loop closure, or more overlap between non-adjacent frames so that long-range constraints exist.
- **High residuals concentrated on one object. It moved. Structure from motion assumes a rigid, static scene, and it will silently distort the geometry to accommodate a moving object** rather than reporting a problem — which is why reconstructions of streets with traffic have characteristic smears.
- **Duplicated structure — two copies of the same wall in the model. Repeated texture fooled place recognition into a false loop closure, or prevented a true one. The hardest failure to fix,** because the incorrect constraint is strongly supported.
- **Always render the point cloud from a novel viewpoint and compare against a photograph taken from roughly there. Residual statistics do not reveal a doubled wall or a bent corridor; a picture reveals both immediately** — which is the same 'test the frame, not the numbers' discipline that CSCE 647 insisted on.

Where to see this taught differently

Original material only goes so far — hearing the same idea in another voice is how it sticks. These are free, and they cover this module specifically.

Resource	What it covers for this module
Schönberger & Frahm — Structure-from-Motion Revisited (COLMAP) (free)	The reference pipeline of §1 , with the seed selection and verification details that matter most.
Triggs et al. — Bundle Adjustment: A Modern Synthesis (free)	The §2 reference. Long, thorough, and the source for the sparsity and gauge-fixing treatment.
Ceres Solver documentation — bundle adjustment example (free)	A working sparse bundle adjustment you can read and run , with the Schur complement option exposed.
Cadena et al. — Past, Present, and Future of SLAM (free survey)	The §3 material in its SLAM form , including loop closure and place recognition.

Exercises

Do these in code, not on paper, unless the exercise says otherwise. The module is not finished until they run.

- 1 **Run COLMAP on 50 of your own photographs** and inspect the reported statistics.
- 2 **Implement the pipeline's steps 1 to 5 yourself** and compare the seed pair you choose against COLMAP's.
- 3 **Deliberately seed from a near-planar pair** and report what happens.

- 4 **Implement PnP with P3P inside RANSAC** and verify against OpenCV.
- 5 **Implement bundle adjustment for 10 cameras** using a sparse solver, and report the error before and after.
- 6 **Run it with squared loss and with Huber** on data containing deliberate mismatches, and compare.
- 7 **Omit gauge fixing** and report what the solver does.
- 8 **Capture a loop** — walk a circuit and return — and measure the closure error without loop closure.
- 9 **Add the loop constraint** and report the improvement across the trajectory.
- 10 **Reconstruct a scene containing a moving object** and locate the distortion it causes.

Self-check

Answer these in writing, from memory, without looking back at the notes. If you cannot, you have read this module but not yet learned it — that is useful information, not a failure.

- 1 Give the seven steps of the incremental pipeline.
- 2 Why is geometric verification what makes it robust?
- 3 Why does the seed pair matter so much?
- 4 What is PnP, and where is it used at 60 Hz?
- 5 State the bundle adjustment objective and explain why it is tractable.
- 6 Name its four practical requirements.
- 7 What is drift, why do residuals not reveal it, and what fixes it?
- 8 Compare incremental and global approaches.
- 9 Name five failure symptoms and their causes.